BABU BANARASI DAS UNIVERSITY

LUCKNOW

Case Study

ON

Project Title: Movie Rating Analysis

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PROJECT

Project Overview & Architecture

The architecture :

- *This project utilizes a standard Big Data pipeline powered by the Hadoop ecosystem to handle large-scale movie datasets.*

Data Flow:

The sequence follows:

Raw Dataset → HDFS → MapReduce Processing → Processed Data → Hive Analysis → Insights

Detailed Explanation of Steps

- **HDFS Storage:** *The raw* ***movie data(Netflix_titles.csv)*** *is first uploaded to the* ***Hadoop Distributed File System (HDFS)*** *for secure, distributed storage.*
- **MapReduce** *Processing: You will use Python with Hadoop Streaming to perform transformations.*
- **Output Storage:** *The results from the Python scripts are stored back in an HDFS output directory.*
- **Hive Analysis:** *Hive is used to create tables over the processed data and run SQL-like queries to extract insights, such as identifying the highest-rated films.*

✓ **Tools Used:**

- *Hadoop Distributed File System* ***(HDFS)***
- *MapReduce* ***(Hadoop Streaming using Python)***
- *Hive* ***(for data analysis)***
- *ChatGPT* ***(Generative AI tool)***

✓ **Technologies:**

- *Python* ***(for mapper and reducer)***
- *CSV dataset*
- *Hadoop Ecosystem*

✓ **Use of Generative AI in the Project:**

For your report, you must document how you integrated AI into your workflow:

- **Code Generation:** *Used AI to write the Python Mapper and Reducer logic.*
- **Query Assistance:** *Used AI to generate complex Hive queries for data aggregation.*
- **Debugging:** *Used AI to resolve specific errors encountered during execution in the Cloudera environment.*

- **Prompt Engineering:** *Refined prompts to ensure the AI-generated code was compatible with the specific CSV structure of the movie dataset.*

Areas Where AI Was Used:

- **Code Generation:** *I used AI to generate the initial Python scripts for the Mapper and Reducer to ensure they were correctly formatted for Hadoop Streaming.*
- **Debugging:** *Whenever I encountered errors in the Cloudera environment or script execution, I used AI to identify the root cause and find a fix.*
- **Query Writing:** *AI helped me write and optimize Hive queries.*
- **Concept Understanding:** *It was used as a learning tool to better understand the HDFS architecture and the MapReduce workflow*

Benefits of Using AI:

- *Faster development*
- *Reduced coding errors*
- *Better understanding of concepts*
- *Improved efficiency*

Validation:

- All AI-generated *code was manually tested and verified within the* ***Cloudera VM*** *to ensure it produced the correct analytical results before final execution.*

Dataset :

- *Comprehensive movie/show metadata repository*
- *Global content production tracking*
- *Records movie and season durations*
- *Brief textual content descriptions included*
- *Unique show identification numbers assigned*

Dataset Structure:

- **The dataset consists of the following columns:**

Column	Description
show_id	*Unique ID*
type	*Movie / TV Show*
title	*Name of movie*
director	*Director name*
cast	*Actors*
country	*Country*
date_added	*Date added*

release_year	*Release year*
rating	*Content rating*
duration	*Movie duration*
listed_in	*Genre*

Dataset Source:

The dataset is a combination of real-world movie data and custom entries created for project demonstration.

Conclusion:

- **In this project, a complete Big Data pipeline was successfully implemented using :**
  HDFS *for storage*, ***MapReduce*** *for counting rating*, ***Hive*** *for* ***SQL*** *analysis*
- *The project helped in understanding the working of distributed data processing and data analysis techniques.*
- *Generative AI acted as a smart assistant throughout the project by helping in code generation, debugging, and understanding complex concepts.*

HDFS Implementation

Step-1: *Upload a file(**Netflix_title.csv**) in the Cloudera terminal and then create a folder in terminal with name "**netflix_project**".*

```
[cloudera@quickstart ~]$ sudo VBoxControl sharedfolder list
Oracle VM VirtualBox Guest Additions Command Line Management Interface Version 5.0.14
(C) 2008-2016 Oracle Corporation
All rights reserved.

Shared Folder mappings (1):

01 - cloudera
[cloudera@quickstart ~]$ sudo mkdir /mnt/shared
[cloudera@quickstart ~]$ sudo mount -t vboxsf cloudera /mnt/shared
[cloudera@quickstart ~]$ cd /mnt/shared
```

```
[cloudera@quickstart shared]$ ls
netflix_titles.csv
[cloudera@quickstart shared]$ cp netflix_titles.csv /home/cloudera/
[cloudera@quickstart shared]$ cd /home/cloudera
[cloudera@quickstart ~]$ ls
bbdu2.txt  bigdata_demo      cm_api.py  Documents  eclipse                    express-deployment.json  lib             mike2.txt  muskan
bbdu4.txt  cloudera-manager  Desktop    Downloads  enterprise-deployment.json  kerberos                 mapreduce_demo  Music      netflix_titles.csv
[cloudera@quickstart ~]$ hdfs dfs -mkdir /netflix_project
[cloudera@quickstart ~]$ hdfs dfs -put netflix_titles.csv /netflix_project
```

MapReduce Implementation

Step-1: Go to the home directory and the checked Werther the dataset exists or not.

```
[cloudera@quickstart ~]$ cd /home/cloudera
[cloudera@quickstart ~]$ ls
bigdata_arslan  cloudera-manager  Desktop    Downloads  enterprise-deployment.json  kerberos  mapper.py       Music   Naincy1.txt    netflix_titles.csv  Pictures  student.txt  Videos
bigdata_demo    cm_api.py         Documents  eclipse    express-deployment.json     lib       mapreduce_demo  nancy   Naincy_folder  parcels             Public    Templates    workspace
[cloudera@quickstart ~]$ head netflix_titles.csv
show_id,type,title,director,cast,country,date_added,release_year,rating,duration,listed_in,description,,,,,,,,,,,,,
s1,Movie,Dick Johnson Is Dead,Kirsten Johnson,,United States,"September 25, 2021",2020,PG-13,90 min,Documentaries,"As her father nears the end of his life, filmmaker Kirsten Johnson stages his death in inventive and comical ways to help
them both face the inevitable.",,,,,,,,,,,,,
s2,TV Show,Blood & Water,,"Ama Qamata, Khosi Ngema, Gail Mabalane, Thabang Molaba, Dillon Windvogel, Natasha Thahane, Arno Greeff, Xolile Tshabalala, Getmore Sithole, Cindy Mahlangu, Ryle De Morny, Greteli Fincham, Sello Maake Ka-Ncube,
Odwa Gwanya, Mekaila Mathys, Sandi Schultz, Duane Williams, Shamilla Miller, Patrick Mofokeng",South Africa,"September 24, 2021",2021,TV-MA,2 Seasons,"International TV Shows, TV Dramas, TV Mysteries","After crossing paths at a party, a C
ape Town teen sets out to prove whether a private-school swimming star is her sister who was abducted at birth.",,,,,,,,,,,,,
s3,TV Show,Ganglands,Julien Leclercq,"Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabiha Akkari, Sofia Lesaffre, Salim Kechiouche, Noureddine Farihi, Geert Van Rampelberg, Bakary Diombera",,"September 24, 2021",2021,TV-MA,1 Season,"Crime T
V Shows, International TV Shows, TV Action & Adventure","To protect his family from a powerful drug lord, skilled thief Mehdi and his expert team of robbers are pulled into a violent and deadly turf war.",,,,,,,,,,,,,
s4,TV Show,Jailbirds New Orleans,,,,"September 24, 2021",2021,TV-MA,1 Season,"Docuseries, Reality TV","Feuds, flirtations and toilet talk go down among the incarcerated women at the Orleans Justice Center in New Orleans on this gritty re
ality series.",,,,,,,,,,,,,
s5,TV Show,Kota Factory,,"Mayur More, Jitendra Kumar, Ranjan Raj, Alam Khan, Ahsaas Channa, Revathi Pillai, Urvi Singh, Arun Kumar",India,"September 24, 2021",2021,TV-MA,2 Seasons,"International TV Shows, Romantic TV Shows, TV Comedies",
"In a city of coaching centers known to train India's finest collegiate minds, an earnest but unexceptional student and his friends navigate campus life.",,,,,,,,,,,,,
s6,TV Show,Midnight Mass,Mike Flanagan,"Kate Siegel, Zach Gilford, Hamish Linklater, Henry Thomas, Kristin Lehman, Samantha Sloyan, Igby Rigney, Rahul Kohli, Annarah Cymone, Annabeth Gish, Alex Essoe, Rahul Abburi, Matt Biedel, Michael T
rucco, Crystal Balint, Louis Oliver",,"September 24, 2021",2021,TV-MA,1 Season,"TV Dramas, TV Horror, TV Mysteries","The arrival of a charismatic young priest brings glorious miracles, ominous mysteries and renewed religious fervor to a
dying town desperate to believe.",,,,,,,,,,,,,
s7,Movie,My Little Pony: A New Generation,"Robert Cullen, José Luis Ucha","Vanessa Hudgens, Kimiko Glenn, James Marsden, Sofia Carson, Liza Koshy, Ken Jeong, Elizabeth Perkins, Jane Krakowski, Michael McKean, Phil LaMarr",,"September 24,
 2021",2021,PG,91 min,Children & Family Movies,"Equestria's divided. But a bright-eyed hero believes Earth Ponies, Pegasi and Unicorns should be pals — and, hoof to heart, she's determined to prove it.",,,,,,,,,,,,,
s8,Movie,Sankofa,Haile Gerima,"Kofi Ghanaba, Oyafunmike Ogunlano, Alexandra Duah, Nick Medley, Mutabaruka, Afemo Omilami, Reggie Carter, Mzuri","United States, Ghana, Burkina Faso, United Kingdom, Germany, Ethiopia","September 24, 2021",
1993,TV-MA,125 min,"Dramas, Independent Movies, International Movies","On a photo shoot in Ghana, an American model slips back in time, becomes enslaved on a plantation and bears witness to the agony of her ancestral past.",,,,,,,,,,,,,
s9,TV Show,The Great British Baking Show,Andy Devonshire,"Mel Giedroyc, Sue Perkins, Mary Berry, Paul Hollywood",United Kingdom,"September 24, 2021",2021,TV-14,9 Seasons,"British TV Shows, Reality TV","A talented batch of amateur bakers
face off in a 10-week competition, whipping up their best dishes in the hopes of being named the U.K.'s best.",,,,,,,,,,,,,
[cloudera@quickstart ~]$ nano mapper.py
```

Step 2: Write a code in the "mapper.py"

```
  GNU nano 2.0.9                                                    File: mapper.py

import sys
import csv

reader = csv.reader(sys.stdin)
next(reader)

for row in reader:
    try:
        rating = row[8]
        if rating:
            print "%s\t1" % rating
    except:
        pass
```

Step 3:Write a code into the “reducer.py” and then give a permission to the file.

```
        pass
[cloudera@quickstart ~]$ nano reducer.py
[cloudera@quickstart ~]$ chmod +x mapper.py reducer.py
```

```
  GNU nano 2.0.9                                          File: reducer.py

import sys
current_rating = None
count = 0
for line in sys.stdin:
    rating, value = line.strip().split("\t")
    value = int(value)
    if current_rating == rating:
        count += value
    else:
        if current_rating:
            print "%s\t%d" % (current_rating, count)
        current_rating = rating
        count = value
if current_rating:
    print "%s\t%d" % (current_rating, count)
```

Step 5: now do local test run

```
[cloudera@quickstart ~]$ cat netflix_titles.csv | python mapper.py | sort | python reducer.py
66 min  1
74 min  1
84 min  1
A       1
G       41
NC-17   3
NR      80
PG      287
PG-13   490
R       799
TV-14   2160
TV-G    220
TV-MA   3208
TV-PG   863
TV-Y    307
TV-Y7   334
TV-Y7-FV        6
UR      3
```

Step 6:Remove the old output folder and the Run MapReduce (Hadoop streaming) on Hadoop

```
[cloudera@quickstart ~]$ hdfs dfs -rm -r /netflix_output
Deleted /netflix_output
[cloudera@quickstart ~]$ hadoop jar /usr/lib/hadoop-mapreduce/hadoop-streaming.jar -file mapper.py -file reducer.py -mapper "python mapper.py" -reducer "python reducer.py" -input /netflix_project/netflix_titles.csv -output /netflix_outpu
t
26/04/21 23:43:13 WARN streaming.StreamJob: -file option is deprecated, please use generic option -files instead.
packageJobJar: [mapper.py, reducer.py] [/usr/lib/hadoop-mapreduce/hadoop-streaming-2.6.0-cdh5.13.0.jar] /tmp/streamjob1903394165295209867.jar tmpDir=null
26/04/21 23:43:16 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
26/04/21 23:43:16 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
26/04/21 23:43:17 WARN hdfs.DFSClient: Caught exception
java.lang.InterruptedException
        at java.lang.Object.wait(Native Method)
        at java.lang.Thread.join(Thread.java:1281)
        at java.lang.Thread.join(Thread.java:1355)
        at org.apache.hadoop.hdfs.DFSOutputStream$DataStreamer.closeResponder(DFSOutputStream.java:967)
        at org.apache.hadoop.hdfs.DFSOutputStream$DataStreamer.endBlock(DFSOutputStream.java:705)
        at org.apache.hadoop.hdfs.DFSOutputStream$DataStreamer.run(DFSOutputStream.java:894)
26/04/21 23:43:17 INFO mapred.FileInputFormat: Total input paths to process : 1
26/04/21 23:43:17 WARN hdfs.DFSClient: Caught exception
java.lang.InterruptedException
        at java.lang.Object.wait(Native Method)
        at java.lang.Thread.join(Thread.java:1281)
        at java.lang.Thread.join(Thread.java:1355)
        at org.apache.hadoop.hdfs.DFSOutputStream$DataStreamer.closeResponder(DFSOutputStream.java:967)
        at org.apache.hadoop.hdfs.DFSOutputStream$DataStreamer.endBlock(DFSOutputStream.java:705)
        at org.apache.hadoop.hdfs.DFSOutputStream$DataStreamer.run(DFSOutputStream.java:894)
26/04/21 23:43:17 INFO mapreduce.JobSubmitter: number of splits:2
26/04/21 23:43:18 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1776831778032_0002
26/04/21 23:43:18 INFO impl.YarnClientImpl: Submitted application application_1776831778032_0002
26/04/21 23:43:18 INFO mapreduce.Job: The url to track the job: http://quickstart.cloudera:8088/proxy/application_1776831778032_0002/
26/04/21 23:43:18 INFO mapreduce.Job: Running job: job_1776831778032_0002
26/04/21 23:43:28 INFO mapreduce.Job: Job job_1776831778032_0002 running in uber mode : false
26/04/21 23:43:28 INFO mapreduce.Job:  map 0% reduce 0%
26/04/21 23:43:42 INFO mapreduce.Job:  map 50% reduce 0%
26/04/21 23:43:44 INFO mapreduce.Job:  map 100% reduce 0%
26/04/21 23:43:52 INFO mapreduce.Job:  map 100% reduce 100%
26/04/21 23:43:53 INFO mapreduce.Job: Job job_1776831778032_0002 completed successfully
```

Step 7: Check the output folder and view final output

```
26/04/21 23:43:54 INFO streaming.StreamJob: Output directory: /netflix_output
[cloudera@quickstart ~]$ hdfs dfs -ls /netflix_output
Found 2 items
-rw-r--r--   3 cloudera supergroup          0 2026-04-21 23:43 /netflix_output/_SUCCESS
-rw-r--r--   3 cloudera supergroup        149 2026-04-21 23:43 /netflix_output/part-00000
[cloudera@quickstart ~]$ hdfs dfs -ls /netflix_output/part-00000
-rw-r--r--   3 cloudera supergroup        149 2026-04-21 23:43 /netflix_output/part-00000
[cloudera@quickstart ~]$ hdfs dfs -cat /netflix_output/part-00000
66 min  1
74 min  1
84 min  1
A       1
G       41
NC-17   3
NR      80
PG      287
PG-13   490
R       799
TV-14   2160
TV-G    220
TV-MA   3207
TV-PG   863
TV-Y    307
TV-Y7   334
TV-Y7-FV        6
UR      3
```

Step 8:

- **Find out the maximum count.**
- **Find out the minimum count.**
- **Find out the 5-top ratings**

```
[cloudera@quickstart ~]$ hdfs dfs -cat /netflix_output/part-00000 | sort -k2 -nr | head -1
TV-MA   3207
[cloudera@quickstart ~]$ hdfs dfs -cat /netflix_output/part-00000 | sort -k2 -n | head -1
66 min  1
[cloudera@quickstart ~]$ hdfs dfs -cat /netflix_output/part-00000 | sort -k2 -nr | head -5
TV-MA   3207
TV-14   2160
TV-PG   863
R       799
PG-13   490
```

Hive Implementation

Step-1: create the database + table and then load the data into the Hive.

```
[cloudera@quickstart ~]$ hive

Logging initialized using configuration in file:/etc/hive/conf.dist/hive-log4j.properties
WARNING: Hive CLI is deprecated and migration to Beeline is recommended.
hive> CREATE DATABASE netflixdb;
OK
Time taken: 1.891 seconds
hive> USE netflixdb;
OK
Time taken: 0.119 seconds
hive> CREATE TABLE rating_count (rating STRING,total INT) ROW FORMAT DELIMITED FIELDS TERMINATED BY '\t';
OK
Time taken: 0.488 seconds
hive> LOAD DATA INPATH '/netflix_output/part-00000' INTO TABLE rating_count;
Loading data to table netflixdb.rating_count
Table netflixdb.rating_count stats: [numFiles=1, totalSize=149]
OK
Time taken: 0.619 seconds
```

Step-2: Check the table

```
hive> SELECT * FROM rating_count;
OK
66 min  1
74 min  1
84 min  1
A       1
G       41
NC-17   3
NR      80
PG      287
PG-13   490
R       799
TV-14   2160
TV-G    220
TV-MA   3207
TV-PG   863
TV-Y    307
TV-Y7   334
TV-Y7-FV        6
UR      3
Time taken: 0.431 seconds, Fetched: 18 row(s)
```

Step 3: find out the highest rating category.

```
hive> SELECT * FROM rating_count ORDER BY total DESC LIMIT 5;
Query ID = cloudera_20260422001818_d9c59e30-723b-47ea-a0e4-8070d8f8b34e
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks determined at compile time: 1
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1  Reduce: 1   Cumulative CPU: 2.0 sec   HDFS Read: 6940 HDFS Write: 48 SUCCESS
Total MapReduce CPU Time Spent: 2 seconds 0 msec
OK
TV-MA   3207
TV-14   2160
TV-PG   863
R       799
PG-13   490
Time taken: 24.68 seconds, Fetched: 5 row(s)
```

Step 4: find out the lowest rating category and filter ratings above 500.

```
hive> SELECT * FROM rating_count ORDER BY total ASC LIMIT 5;
Query ID = cloudera_20260422001919_5db9070f-e98e-4b0e-b67f-9a6077f7c9ce
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks determined at compile time: 1
```

```
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1  Reduce: 1   Cumulative CPU: 1.9 sec   HDFS Read: 7508 HDFS Write: 3 SUCCESS
Total MapReduce CPU Time Spent: 1 seconds 900 msec
OK
18
Time taken: 22.033 seconds, Fetched: 1 row(s)
hive> SELECT * FROM rating_count WHERE total > 500;
OK
R       799
TV-14   2160
TV-MA   3207
TV-PG   863
Time taken: 0.139 seconds, Fetched: 4 row(s)
```

Step 5: find average of rating count.

```
hive> SELECT AVG(total) FROM rating_count;
Query ID = cloudera_20260422002222_cb432737-fe91-46d5-849a-8b0a8df0ad34
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks determined at compile time: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1776831778032_0006, Tracking URL = http://quickstart.cloudera:8088/proxy/application_1776831778032_0006/
Kill Command = /usr/lib/hadoop/bin/hadoop job  -kill job_1776831778032_0006
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 1
2026-04-22 00:22:34,800 Stage-1 map = 0%,  reduce = 0%
2026-04-22 00:22:40,266 Stage-1 map = 100%,  reduce = 0%, Cumulative CPU 0.85 sec
2026-04-22 00:22:47,627 Stage-1 map = 100%,  reduce = 100%, Cumulative CPU 1.92 sec
MapReduce Total cumulative CPU time: 1 seconds 920 msec
Ended Job = job_1776831778032_0006
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1  Reduce: 1   Cumulative CPU: 1.92 sec   HDFS Read: 8023 HDFS Write: 18 SUCCESS
Total MapReduce CPU Time Spent: 1 seconds 920 msec
OK
489.1111111111111
Time taken: 22.079 seconds, Fetched: 1 row(s)
```